

MEENATCHI K V

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Objective

Seeking a responsible job to gain and enrich knowledge and expertise with opportunities for professional challenges and to utilize my analytical and problem-solving skills to the fullest.

Education

Amrita Vishwa Vidyapeetham

PhD. Computer Science and Engineering
9.38 CGPA - Coursework

Currently Pursuing

Anna University

ME. Computer Science and Engineering with Big Data Spln.
9.39 CGPA

Graduation Date

May 2021

Rajalakshmi Engineering College

BE. Computer Science and Engineering
8.49 CGPA

Graduation Date

May 2019

Work Experience

Vel Tech

Project Associate

Chennai, TN

Sep 2021 - Apr 2023

Saveetha Engineering College

Assistant Professor, CSE (Cyber Security)

Chennai, TN

May 2023 - Nov 2023

Institute of Science and Technology, NMV University

Assistant Professor, AIML

Chennai, TN

Dec 2023 - June 2024

Adi Shankara Institute of Engineering and Technology

Assistant Professor, CSE(AI)

Kalady, Kerala

Aug 2024 - June 2026

Academic Experience

Assistant Professor

May 2023 – Present

- Deliver undergraduate courses in AI, ML, and core CSE subjects
- Mentor student projects in AI-based applications and research
- Guided student mini-projects and main-projects
- Subjects - Python, Data Structures, Augmented Reality, Discrete Mathematics
- Served as a resource person in workshops and FDPs related to
 - Python - Numpy, Pandas, Opencv
 - Generative AI - Prompt Engineering and Other AI Tools
 - Workflow Automation using n8n

- Led project teams and coordinated multi-member research activities
- Trained new recruits and standardized onboarding processes
- Worked on “Drone Assisted Mapping of Seaweeds of Odisha Coast including Chilika Lake for Gracilaria sp. and Enteromorpha sp. and development of Value-added seaweed projects”

Project Experience

A Unified Physics-Informed Diffusion Framework for MRI Reconstruction and Synthetic MRI Generation

- Conducted a systematic review with experiential validation and benchmarking of MRI Reconstruction techniques
- Conducted Statistical Validation of the existing techniques.

Reconstruction of Murals using Tri-Cycle GAN

- Designed a Tri-Cycle GAN architecture using TensorFlow & Keras
- Trained on a custom dataset (300+ temple mural images)
- Achieved 80% restoration success rate in color stabilization
- Improved accuracy by 10% over SE-ResNet GAN baseline
- Surpassed benchmark precision metrics by 27%.

Phonic Eye for the Visually Impaired

- Developed real-time object detection system using TensorFlow & Keras
- Achieved 140 FPS inference speed via Flask + TensorFlow Serving
- Built Android application with text-to-speech scene narration (<6 sec latency)
- Reduced manual object identification effort by 60%

Shock Proof Augmented Reality Switches

- Built AR-based IoT control system using Unity
- Integrated mobile AR interface with hardware automation
- Achieved 100% functional success in prototype testing

Publications

Phonic Eye for the Visually Impaired, International Journal of Engineering and Advanced Technology (IJEAT), 2019

Enhancing Disaster Vulnerability Analysis and Plot Price Prediction in Local Areas with Bayesian-Optimized Random Forest, XGBoost, and LSTM Models Using TPE, Conference Proceedings at IEEE ACCESS '25, 2025

AirCo: Real-Time Air Quality Prediction and Data Analysis, Conference Proceedings at ICIMRBE, 2025

Research Interests

Machine Learning, Deep Learning, Computer Vision, Generative AI, IoT Systems

Technical Skills

Programming	Python, C, C++, Java
Frameworks & Tools	Tensorflow, Keras, Flask, Tensorflow Serving, XAMPP, Unity, Android Studio
Web Technologies	HTML, CSS, SQL

Soft Skills

Analytical Thinking, Mentorship, Communication, Problem Solving, Team Leadership

I hereby declare that the above details are accurate to the best of my knowledge.

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